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Artificial Intelligence & Machine Learning - Task 2 Report

Feature Engineering, Model Optimization & Performance Comparison

1. Executive Summary

This project demonstrates an end-to-end machine learning pipeline using the **California Housing Dataset**. The objective is to build, optimize, and compare multiple regression algorithms to predict median house values accurately. Proper data preprocessing, feature scaling, and comparative performance metrics were applied to select the optimal model.

Tab 2

2. Methodology & Workflow

Dataset Loading: Utilized the California Housing dataset containing numeric attributes such as Median Income, House Age, Average Rooms, and geographic coordinates.

Data Preprocessing: Checked for null values, split the dataset into 80% training and 20% testing sets, and applied `StandardScaler` to normalize feature distributions.

Model Training: Implemented four distinct algorithms:

1. Linear Regression (Baseline)
2. Decision Tree Regressor
3. Random Forest Regressor
4. Gradient Boosting Regressor

Key Insights & Conclusion

- **Ensemble Superiority:** Ensemble methods (**Random Forest** and **Gradient Boosting**) drastically outperformed traditional linear regression and individual decision trees by capturing complex non-linear relationships.
- **Best Model Selection:** The **Random Forest Regressor** achieved the highest R^2 Score (~ 0.805), explaining roughly 80.5% of the variance in housing prices.
- **Deliverables Check:** The trained Random Forest model has been successfully serialized and saved using `joblib` (`best_house_price_model.pkl`) for future deployment.

Prepared by Shivam